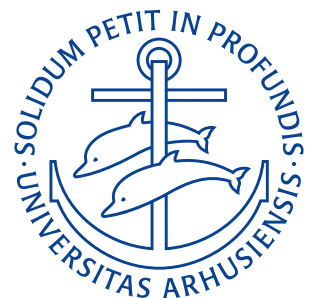


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Homework Exercises

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Lecture 1: Introduction

26. Januar 2026

Problem 1.1a

Show that

$$\delta_{ij}\delta_{ij} = 3.$$

DerivationExpanding $\delta_{ij}\delta_{ij}$ we get

$$\delta_{ij}\delta_{ij} = \delta_{11}\delta_{11} + \delta_{12}\delta_{12} + \delta_{13}\delta_{13} + \delta_{21}\delta_{21} + \delta_{22}\delta_{22} + \delta_{23}\delta_{23} + \delta_{31}\delta_{31} + \delta_{32}\delta_{32} + \delta_{33}\delta_{33}.$$

As $\delta_{11} = \delta_{22} = \delta_{33} = 1$ the above becomes:

$$\delta_{ij}\delta_{ij} = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 3.$$

$\delta_{ij}\delta_{ij} = 3$

RESULT

Problem 1.1c

Show that

$$\alpha_{ij}b_{jk} = \alpha_{in}b_{nk}.$$

Derivation

We expand by summing over the common index j :

$$\alpha_{ij}b_{jk} = \sum_j \alpha_{ij}b_{jk} = \alpha_{i1}b_{1k} + \alpha_{i2}b_{2k} + \alpha_{i3}b_{3k}.$$

Now we sum over n instead:

$$\alpha_{i1}b_{1k} + \alpha_{i2}b_{2k} + \alpha_{i3}b_{3k} = \sum_n \alpha_{in}b_{nk}.$$

Q.E.D.

$\alpha_{ij}b_{jk} = \alpha_{in}b_{nk}$

RESULT

Problem 1.1e

Show that

$$\delta_{ij}C_{jkl} = C_{ikl}.$$

Derivation

Expanding we get

$$\delta_{ij}C_{jkl} = \delta_{11}C_{1kl} + \delta_{22}C_{2kl} + \delta_{33}C_{3kl} = C_{ikl}.$$

$\delta_{ij}C_{jkl} = C_{ikl}$

RESULT

Lecture 2: Traction, Stress, and Equilibrium

2. Februar 2026

Lecture 3: The Cauchy Theorem & Principal Stresses

9. Februar 2026

Problem 2.1

Find the components of the traction on a plane defined by $n_1 = 1/\sqrt{2}$, $n_2 = 1/\sqrt{2}$, and $n_3 = 0$ for the following states of stress:

- (a) $\sigma_{11} = \sigma, \sigma_{12} = 0, \sigma_{13} = 0, \sigma_{21} = \sigma, \sigma_{22} = 0, \sigma_{33} = \sigma.$
 (b) $\sigma_{11} = \sigma, \sigma_{12} = \sigma, \sigma_{13} = 0, \sigma_{22} = \sigma, \sigma_{23} = 0, \sigma_{33} = 0.$

Derivation

- (a) The stress state in (a) is

$$\sigma = \begin{bmatrix} \sigma & 0 & 0 \\ \sigma & 0 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma \end{bmatrix}.$$

The Cauchy Theorem states that

$$T_i = \sigma_{ji} n_j \implies T = \sigma \cdot n.$$

Our plane is:

$$n = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}.$$

Hence,

$$T = \begin{bmatrix} \sigma & 0 & 0 \\ \sigma & 0 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{pmatrix} \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{pmatrix}.$$

- (b) For (b) our stress state is:

$$\sigma = \begin{bmatrix} \sigma & \sigma & 0 \\ \sigma_{21} & \sigma & 0 \\ \sigma_{31} & \sigma_{32} & 0 \end{bmatrix},$$

so

$$T = \begin{bmatrix} \sigma & \sigma & 0 \\ \sigma_{21} & \sigma & 0 \\ \sigma_{31} & \sigma_{32} & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{pmatrix} \sqrt{2}\sigma \\ \frac{\sigma}{\sqrt{2}} + \frac{\sigma_{21}}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{pmatrix}.$$

$$\begin{aligned} \text{(a)} \quad T &= \begin{bmatrix} \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{bmatrix} \\ \text{(b)} \quad T &= \begin{bmatrix} \sqrt{2}\sigma \\ \frac{\sigma}{\sqrt{2}} + \frac{\sigma_{21}}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{bmatrix}. \end{aligned}$$

RESULT

Problem 2.6

At a given point in a body,

$$\sigma_{11} = \sigma_{22} = \sigma_{33} = -p,$$

$$\sigma_{12} = \sigma_{23} = \sigma_{31} = 0.$$

Show that the normal stresses are equal to $-p$ and that the shearing stresses vanish for any other Cartesian coordinate system.

Derivation

Here, our stress tensor is

$$\boldsymbol{\sigma} = \begin{bmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{bmatrix} = -p\mathbf{I}.$$

Using the Cauchy theorem $\mathbf{T} = \boldsymbol{\sigma} \cdot \mathbf{n}$ we get:

$$\mathbf{T} = (-p\mathbf{I}) \cdot \mathbf{n} = -p\mathbf{n}.$$

Hence, $\mathbf{T}$ is parallel to $\mathbf{n}$ meaning the traction does not have any components tangent to the plane and hence the stress is entirely normal with size $-p$.

Lecture 4: Equilibrium Equations & Principle of Angular Momentum

16. Februar 2026

Problem 2.2

The state of stress at a point P in a structure is given by

$$\sigma_{11} = 20000,$$

$$\sigma_{22} = -15000,$$

$$\sigma_{33} = 3000$$

$$\sigma_{12} = 2000,$$

$$\sigma_{23} = 2000,$$

$$\sigma_{31} = 1000.$$

- Compute the scalar components T_1 , T_2 , and T_3 of the traction $\mathbf{T}$ on the plane passing through P whose outward normal vector $\mathbf{n}$ makes equal angles with the coordinate axes x_1 , x_2 , x_3 .
- Compute the normal and tangential components of stress on this plane.
- Determine the principal stresses.
- Determine the direction of maximum principal stress.

Derivation

- We are given the following stress state:

$$\boldsymbol{\sigma} = \begin{bmatrix} 20000 & 2000 & \cdot \\ \cdot & -15000 & 2000 \\ 1000 & \cdot & 3000 \end{bmatrix}.$$

We assume this is symmetric i.e. it satisfies the standard case of $\sigma_{ij} = \sigma_{ji}$ and we therefore get:

$$\boldsymbol{\sigma} = \begin{bmatrix} 20000 & 2000 & 1000 \\ 2000 & -15000 & 2000 \\ 1000 & 2000 & 3000 \end{bmatrix}.$$

As $\mathbf{n}$ makes equal angles with x_1 , x_2 , and x_3 , $\mathbf{n}$ must be:

$$\mathbf{n} = \frac{1}{\sqrt{3}}(1, 1, 1).$$

The traction vector on the plane is:

$$\mathbf{T} = \boldsymbol{\sigma} \mathbf{n}$$

so,

$$\begin{aligned} T_1 &= \frac{1}{\sqrt{3}} (20000 + 2000 + 1000) = \frac{23000}{\sqrt{3}} \\ T_2 &= \frac{1}{\sqrt{3}} (2000 - 15000 + 2000) = -\frac{11000}{\sqrt{3}} \\ T_3 &= \frac{1}{\sqrt{3}} (1000 + 2000 + 3000) = \frac{6000}{\sqrt{3}}. \end{aligned}$$

(b) The normal stress is given as

$$\sigma_{nn} = \mathbf{T}_n \cdot \mathbf{n}.$$

I.e.

$$\begin{aligned} \sigma_{nn} &= \frac{1}{\sqrt{3}} (T_1 + T_2 + T_3) \\ &= \frac{1}{3} (23000 - 11000 + 6000) \\ &= 6000. \end{aligned}$$

Hence, the normal traction vector $\mathbf{N}$ is

$$\mathbf{N} = \sigma_{nn} \cdot \mathbf{n} = 6000 \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

As the total traction $\mathbf{T}$ is the algebraic sum of the normal traction $\mathbf{N}$ and the tangential traction $\mathbf{S}$ we get:

$$\mathbf{T} = \mathbf{N} + \mathbf{S} \implies \mathbf{S} = \mathbf{T} - \mathbf{N}.$$

So

$$\mathbf{S} = \frac{1}{\sqrt{3}} \begin{pmatrix} 23000 \\ -11000 \\ 6000 \end{pmatrix} - \frac{6000}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 17000 \\ -17000 \\ 0 \end{pmatrix}.$$

(d) For principal stresses we have the condition:

$$|\sigma_{ji} - \lambda \delta_{ij}| = 0$$

i.e.:

$$\begin{vmatrix} 20000 - \lambda & 2000 & 1000 \\ 2000 & -15000 - \lambda & 2000 \\ 1000 & 2000 & 3000 - \lambda \end{vmatrix} = 0 \rightarrow \lambda_1 \rightarrow -15319,7, \quad \lambda_2 \rightarrow 3133,4, \quad \lambda_3 \rightarrow 20186,3.$$

Therefore our principal stresses are:

$$\sigma^{(1)} = -15319,7$$

$$\sigma^{(2)} = 3133,4$$

$$\sigma^{(3)} = 20186,3.$$

(e) For the maximum principal stress $\lambda_3 = 20186,3$ we solve

$$(\sigma_{ji} - \lambda \delta_{ij}) n_j = 0.$$

I.e.

$$\begin{pmatrix} 20000 - 20186,3 & 2000 & 1000 \\ 2000 & -15000 - 20186,3 & 2000 \\ 1000 & 2000 & 3000 - 20186,3 \end{pmatrix} \cdot \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Solving this gives $n_1 = 15,3299$, $n_2 = 0,928198$, $n_3 = 1$. Now we scale this by the length to get a unit vector:

$$\mathbf{n}_{\max} = \frac{\begin{pmatrix} 15,3299 \\ 0,928198 \\ 1 \end{pmatrix}}{\sqrt{15,3299^2 + 0,928198^2 + 1^2}} = \begin{pmatrix} 0,996 \\ 0,0603 \\ 0,0650 \end{pmatrix}.$$

(a)	$\mathbf{T} = \frac{1}{\sqrt{3}} \begin{pmatrix} 23000 \\ -11000 \\ 6000 \end{pmatrix}$
(b)	$\mathbf{N} = \frac{1}{\sqrt{3}} \begin{pmatrix} 6000 \\ 6000 \\ 6000 \end{pmatrix}$
	$\mathbf{S} = \frac{1}{\sqrt{3}} \begin{pmatrix} 17000 \\ -17000 \\ 0 \end{pmatrix}$
(d)	$\sigma^{(1)} = -15319,7$ $\sigma^{(2)} = 3133,4$ $\sigma^{(3)} = 20186,3$
(e)	$\mathbf{n}_{\max} = \begin{pmatrix} 0,996 \\ 0,060 \\ 0,065 \end{pmatrix}.$

RESULT

Problem 2.3

The state of stress at a point is given by the components of stress tensor σ_{ij} . A plane is defined by the direction cosines of the normal $(1/2, 1/2, 1/\sqrt{2})$. State the general conditions for which the traction on the plane has the same direction as the x_2 -axis and a magnitude of 1,0.

Derivation

We use

$$T_i = \sigma_{ji} n_j = \boldsymbol{\sigma} \cdot \mathbf{n}.$$

We are given

$$\mathbf{n} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

The traction should be in the x_2 axis and have magnitude 1,0 so it must be

$$\mathbf{T} = \begin{pmatrix} 0 \\ 1,0 \\ 0 \end{pmatrix} = \mathbf{e}_2.$$

So the condition is simply

$$\boldsymbol{\sigma} \mathbf{n} = \mathbf{e}_2.$$

$\boldsymbol{\sigma} \mathbf{n} = \mathbf{e}_2$

RESULT

Problem 2.4

Determine the body forces for which the following stress field describes a state of equilibrium

$$\sigma_{xx} = -2x^2 - 3y^2 - 5z,$$

$$\sigma_{xy} = x + 4xy - 6,$$

$$\sigma_{yy} = -2y^2 + 7,$$

$$\sigma_{xz} = -3x + 2y + 1.$$

$$\sigma_{zz} = 4x + y + 3z - 5,$$

$$\sigma_{yz} = 0.$$

Derivation

We have that

$$\sigma_{ij,j} + f_i = 0.$$

which is:

$$\begin{aligned} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} + f_x &= 0 \\ \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} + f_y &= 0 \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + f_z &= 0. \end{aligned}$$

In the x -direction we get

$$\begin{aligned} \frac{\partial(-2x^2 - 3y^2 - 5z)}{\partial x} + \frac{\partial(x + 4xy - 6)}{\partial y} + \frac{\partial(-3x + 2y + 1)}{\partial z} + f_x &= 0 \\ -4x + 4x + f_x &= 0 \\ f_x &= 0. \end{aligned}$$

In the y -direction we get

$$\begin{aligned} \frac{\partial(x + 4xy - 6)}{\partial x} + \frac{\partial(-2y^2 + 7)}{\partial y} + \frac{\partial 0}{\partial z} + f_y &= 0 \\ 1 + 4y - 4y + f_y &= 0 \\ f_y &= -1. \end{aligned}$$

And in the z -direction we get

$$\begin{aligned} \frac{\partial(-3x + 2y + 1)}{\partial x} + \frac{\partial 0}{\partial y} + \frac{\partial(4x + y + 3z - 5)}{\partial z} + f_z &= 0 \\ -3 + 3 + f_z &= 0 \\ f_z &= 0. \end{aligned}$$

$$\mathbf{f} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = -\mathbf{e}_2$$

RESULT

Problem 2.5

Determine whether the following stress field is admissible in an elastic body when body forces are negligible.

$$\boldsymbol{\sigma} = \begin{bmatrix} yz+4 & z^2+2x & 5y+z \\ \cdot & xz+3y & 8x^3 \\ \cdot & \cdot & 2xyz \end{bmatrix}.$$

Derivation

We once again use

$$\sigma_{ij,j} + f_i = 0.$$

With negligible body forces this becomes

$$\sigma_{ij,j} = 0.$$

I.e.

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} = 0$$

$$\frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} = 0$$

$$\frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = 0.$$

In the x -direction we get

$$\frac{\partial(yz+4)}{\partial x} + \frac{\partial(z^2+2x)}{\partial y} + \frac{\partial(5y+z)}{\partial z} = 0$$

$$0 + 0 + 1 \neq 0.$$

Hence, the stress field is not admissible with negligible body forces.

It is not.

RESULT

Problem 2.8

The stress tensor at a point $P(x, y, z)$ in a solid is

$$\boldsymbol{\sigma} = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}.$$

- (a) Show that the principal stresses are 4, 1, and -2 .
- (b) Find the orientation of the principal planes

Derivation

- (a) We have the characteristic equation

$$|\sigma_{ji} - \lambda \delta_{ij}| = 0.$$

I.e.

$$\begin{vmatrix} 3-\lambda & 1 & 1 \\ 1 & 0-\lambda & 2 \\ 1 & 2 & 0-\lambda \end{vmatrix} = 0 \implies \lambda_1 \rightarrow -2, \quad \lambda_2 \rightarrow 1, \quad \lambda_3 \rightarrow 4.$$

- (b) The orientation of the principal planes are the eigenvectors of these. I.e. we have to solve

$$(\sigma_{ji} - \lambda \delta_{ij}) n_j = 0.$$

For each λ .

For $\lambda_1 = -2$ we get

$$\begin{bmatrix} 5 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} \cdot \mathbf{n}_1 = 0 \implies \mathbf{n}_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_1 = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

For $\lambda_2 = 1$ we get

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 2 & -1 \end{bmatrix} \mathbf{n}_2 = 0 \implies \mathbf{n}_2 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_2 = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}.$$

For $\lambda_3 = 4$ we get

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -4 & 2 \\ 1 & 2 & -4 \end{bmatrix} \mathbf{n}_3 = 0 \implies \mathbf{n}_3 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_3 = \begin{pmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}.$$

$$(a) \quad \sigma^{(1)} = -2$$

$$\sigma^{(2)} = 1$$

$$\sigma^{(3)} = 4$$

$$(b) \quad \hat{n}_1 = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\hat{n}_2 = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$\hat{n}_3 = \begin{pmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}.$$

RESULT

Problem 2.9

The stress tensor at a point $P(x, y, z)$ in a solid is

$$\boldsymbol{\sigma} = \begin{bmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{bmatrix}.$$

- (a) Determine the stresses with respect to a set of axes x' , y' , and z' for which the matrix of direction cosines is

$$[\alpha_{ij}] = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (b) Compute the principal stresses using both the x -, y -, and z - and x' -, y' -, and z' -axes as the basis.
 (c) Find the orientation of the principal planes with respect to the x -, y -, and z -axes.

Derivation

- (a) To transform coordinates we use

$$\mathbf{A}' = \mathbf{R} \mathbf{A} \mathbf{A}'^T.$$

Hence

$$\boldsymbol{\sigma}' = \boldsymbol{\alpha} \boldsymbol{\sigma} \boldsymbol{\alpha}^T$$

which gives

$$\boldsymbol{\sigma}' = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{bmatrix} 107,86 & -32,99 & -2,32 \\ -32,99 & 162,14 & 35,98 \\ -2,32 & 35,98 & 100 \end{bmatrix}.$$

- (b) Here we use the characteristic equation from before to find the eigenvalues:

$$\begin{bmatrix} 120 - \lambda & 40 & 30 \\ 40 & 150 - \lambda & 20 \\ 30 & 20 & 100 - \lambda \end{bmatrix} = 0 \implies \lambda_1 = 190,405, \quad \lambda_2 = 102,423, \quad \lambda_3 = 77,1724.$$

and

$$\begin{bmatrix} 107,86 - \lambda & -32,99 & -2,32 \\ -32,99 & 162,14 - \lambda & 35,98 \\ -2,32 & 35,98 & 100 - \lambda \end{bmatrix} \rightarrow \lambda_1 = 190,403, \quad \lambda_2 = 102,424, \quad \lambda_3 = 77,1727.$$

Which confirms that principal stresses are unchanged by rotation.

- (c) Here we solve

$$|\sigma_{ji} - \lambda \delta_{ij}| n_j = 0$$

for each eigenvalue and get:

$$\begin{bmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{bmatrix} \rightarrow \left(\begin{array}{ccc} 190.405 & 102.423 & 77.1724 \\ \{1.61658, 2.09537, 1.\} & \{0.821389, -1.11094, 1.\} & \{-0.911651, 0.226096, 1.\} \end{array} \right).$$

Now we scale each $\mathbf{n}$ and get:

$$\mathbf{n}_1 = \begin{pmatrix} 0,5714 \\ 0,7406 \\ 0,3535 \end{pmatrix}$$

$$\mathbf{n}_2 = \begin{pmatrix} 0,4816 \\ -0,6514 \\ 0,5863 \end{pmatrix}$$

$$\mathbf{n}_3 = \begin{pmatrix} 0,6645 \\ -0,1648 \\ -0,7289 \end{pmatrix}.$$

$$(a) \quad \boldsymbol{\sigma} = \begin{pmatrix} 107,86 & -32,99 & -2,32 \\ -32,99 & 162,14 & 35,98 \\ -2,32 & 35,98 & 100 \end{pmatrix}$$

$$(b) \quad \lambda_1 = 190,405$$

$$\lambda_2 = 102,423$$

$$\lambda_3 = 77,1724$$

$$(c) \quad \mathbf{n}_1 = \begin{pmatrix} 0,5714 & 0,7406 & 0,3535 \end{pmatrix}$$

$$\mathbf{n}_2 = \begin{pmatrix} 0,4816 & -0,6514 & 0,5863 \end{pmatrix}$$

$$\mathbf{n}_3 = \begin{pmatrix} 0,6645 & -0,1648 & -0,7289 \end{pmatrix}.$$

RESULT

Lecture 5: The Strain Tensor

23. Februar 2026

Problem 3.1

The components of a displacement field are

$$\begin{aligned}u_x &= (x^2 + 20) \cdot 10^{-4}, \\u_y &= 2yz \cdot 10^{-3}, \\u_z &= (z^2 - xy) \cdot 10^{-3}.\end{aligned}$$

- (a) Consider two points in the undeformed system, $(2, 5, 7)$ and $(3, 8, 9)$. Find the change in distance between these points.

Derivation

The displacement at $A = (2, 5, 7)$ is

$$\begin{aligned}u_x(A) &= (2^2 + 20) \cdot 10^{-4} = 0,0024 \\u_y(A) &= 2 \cdot 5 \cdot 7 \cdot 10^{-3} = 0,070 \\u_z(A) &= (7^2 - 2 \cdot 5) \cdot 10^{-3} = 0,039\end{aligned}$$

so

$$\mathbf{u}(A) = \begin{pmatrix} 0,0024 \\ 0,070 \\ 0,039 \end{pmatrix}.$$

And at $B = (3, 8, 9)$ it is

$$\begin{aligned}u_x(B) &= (3^2 + 20) \cdot 10^{-4} = 0,0029 \\u_y(B) &= 2 \cdot 8 \cdot 9 \cdot 10^{-3} = 0,144 \\u_z(B) &= (9^2 - 3 \cdot 8) \cdot 10^{-3} = 0,057\end{aligned}$$

so

$$\mathbf{u}(B) = \begin{pmatrix} 0,0029 \\ 0,144 \\ 0,057 \end{pmatrix}.$$

The deformed positions are:

$$\begin{aligned}A' &= A + \mathbf{u}(A) = \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix} + \begin{pmatrix} 0,0024 \\ 0,070 \\ 0,039 \end{pmatrix} = \begin{pmatrix} 2,0024 \\ 5,070 \\ 7,039 \end{pmatrix} \\B' &= B + \mathbf{u}(B) = \begin{pmatrix} 3 \\ 8 \\ 9 \end{pmatrix} + \begin{pmatrix} 0,0029 \\ 0,144 \\ 0,057 \end{pmatrix} = \begin{pmatrix} 3,0029 \\ 8,144 \\ 9,057 \end{pmatrix}.\end{aligned}$$

Our position vectors are:

$$AB = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$AB' = \begin{pmatrix} 1.0005 \\ 3.074 \\ 2.018 \end{pmatrix}.$$

These have the lengths:

$$|AB| = \sqrt{1^2 + 3^2 + 2^2} = 3.74166$$

$$|AB'| = \sqrt{1.0005^2 + 3.074^2 + 2.018^2} = 3.81088.$$

The change in length is then:

$$\Delta L = |AB'| - |AB| = 3.81088 - 3.74166 = 0.06922.$$

$\Delta L = 0.06922$

RESULT